STELLARE SOFTWARE

DOCUMENT ENCRYPTION

Version 6 · User Manual

STANDARD £9	ADVANCED £29	PROFESSIONAL £59

AES-256 · ChaCha20 · ARGON2id · 7-LAYERS · KEY-FILE
Universal File Encryption · Triple-Pass AES + ChaCha20
Argon2id Key Derivation · Per-File Random Salts
Machine-Locked Licensing · Advanced Key File System

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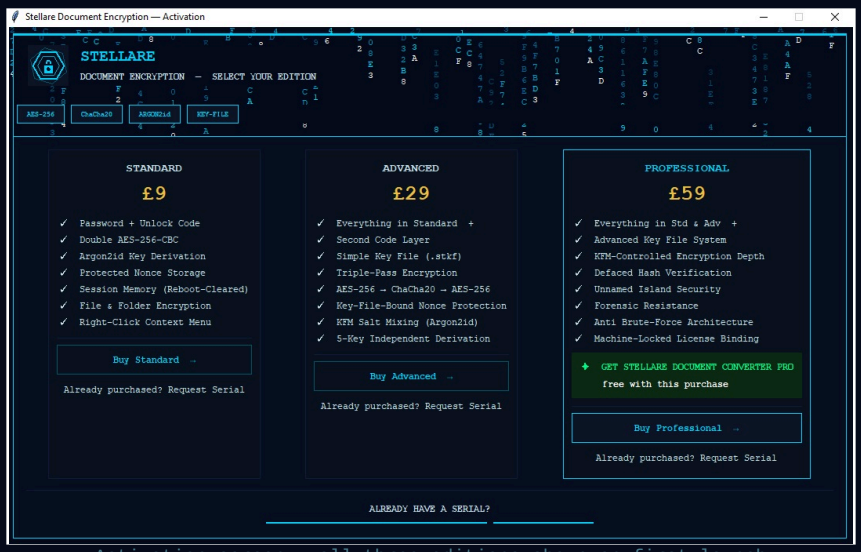
A · Edition Guide

Stellare Document Encryption is available in three editions — Standard, Advanced and Professional. All editions are one-time purchases with no subscription. Each edition is a separate serial that unlocks additional features in the same application.

The activation screen shows all three editions on first launch:



Activation screen — all three editions shown on first launch

STANDARD	ADVANCED	PROFESSIONAL
£9	£29	£59
✓ Password + Unlock Code	✓ Everything in Standard +	✓ Everything in Std & Adv +
✓ Double AES-256-CBC	✓ Second Code Layer	✓ Advanced Key File System
✓ Argon2id Key Derivation	✓ Simple Key File (.stkf)	✓ KFM-Controlled Encryption Depth
✓ Protected Nonce Storage	✓ Triple-Pass Encryption	✓ Defaced Hash Verification
✓ Session Memory (Reboot-Cleared)	✓ AES-256 → ChaCha20 → AES-256	✓ Unnamed Island Security
✓ File & Folder Encryption	✓ Key-File-Bound Nonce Protection	✓ Forensic Resistance
✓ Right-Click Context Menu	✓ KFM Salt Mixing (Argon2id)	✓ Anti Brute-Force Architecture
	✓ 5-Key Independent Derivation	✓ Machine-Locked License Binding
		· FREE: Stellare Document Converter Pro

One download. One application. Your serial key determines which edition is active.

A.1 Standard — £9

Professional-grade encryption for everyday use. Password and Unlock Code protection with Double AES-256-CBC and Argon2id key derivation. Protected nonce storage and session memory that auto-clears on reboot. Right-click context menu integration for fast single-file and folder operations.

- ✓ Password + Unlock Code
- ✓ Double AES-256-CBC
- ✓ Argon2id Key Derivation
- ✓ Protected Nonce Storage
- ✓ Session Memory (Reboot-Cleared)
- ✓ File & Folder Encryption
- ✓ Right-Click Context Menu

A.2 Advanced — £29

The Advanced edition adds a Second Code as a third independent authentication factor and the Simple Key File system. Encryption uses a triple-pass cipher — AES-256 then ChaCha20 then AES-256 — with five independent keys derived from a 160-byte Argon2id output. The KFM salt mixing means the key file actively changes the salt fed to Argon2id — breaking one algorithm reveals only another layer with a completely independent key.

- ✓ Everything in Standard +
- ✓ Second Code Layer
- ✓ Simple Key File (.stkf)
- ✓ Triple-Pass Encryption
- ✓ AES-256 → ChaCha20 → AES-256
- ✓ Key-File-Bound Nonce Protection
- ✓ KFM Salt Mixing (Argon2id)
- ✓ 5-Key Independent Derivation

A.3 Professional — £59

The Professional edition adds the Advanced Key File system — a novel cryptographic construction. KFM-Controlled Encryption Depth means the key file actively determines how many and which type of extra passes are applied — without the correct key file an attacker cannot know the encryption depth. Forensic analysis is unable to confirm the file even contains encrypted data. Designed to counter GPU-accelerated and AI-driven brute force attacks.

- ✓ Everything in Std & Adv +
- ✓ Advanced Key File System
- ✓ KFM-Controlled Encryption Depth
- ✓ Defaced Hash Verification
- ✓ Unnamed Island Security
- ✓ Forensic Resistance
- ✓ Anti Brute-Force Architecture
- ✓ Machine-Locked License Binding

· BONUS INCLUDED WITH PROFESSIONAL EDITION

Stellare Document Converter Pro is included free with every Professional purchase.
After activation, a second automatic email delivers your download link and activation code.

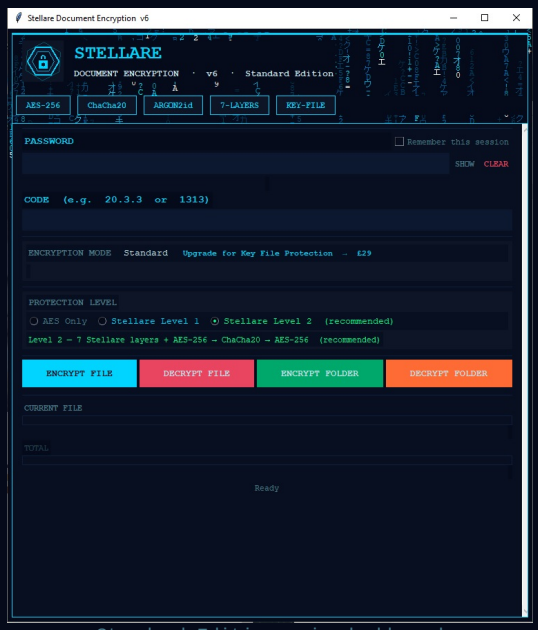
A.4 Feature Comparison

Feature	Standard £9	Advanced £29	Professional £59
One-time purchase	✓	✓	✓
Machine-locked licence	✓	✓	✓
Password + Unlock Code	✓	✓	✓
Double AES-256-CBC	✓	✓	✓
Argon2id Key Derivation	✓	✓	✓
Protected Nonce Storage	✓	✓	✓
Session Memory (Reboot-Cleared)	✓	✓	✓
File & Folder Encryption	✓	✓	✓
Right-Click Context Menu	✓	✓	✓
Second Code Layer	—	✓	✓
Triple-Pass AES+ChaCha20+AES	—	✓	✓
Simple Key File (.stkf)	—	✓	✓
Key-File-Bound Nonce Protection	—	✓	✓
KFM Salt Mixing (Argon2id)	—	✓	✓
5-Key Independent Derivation	—	✓	✓
Advanced Key File System	—	—	✓
KFM-Controlled Encryption Depth	—	—	✓
Defaced Hash Verification	—	—	✓
Unnamed Island Security	—	—	✓
Forensic Resistance	—	—	✓
Anti Brute-Force Architecture	—	—	✓
Stellare Document Converter Pro (FREE)	·	·	·

NOTE Upgrade at any time — purchase the next edition serial and activate on the same machine. Existing files remain accessible with your current credentials.

B.1 · Standard Edition

PASSWORD + CODE · DOUBLE AES-256-CBC · ARGON2ID KEY DERIVATION



Standard Edition main dashboard

STANDARD · £9

- ✓ Password + Unlock Code
- ✓ Double AES-256-CBC
- ✓ Argon2id Key Derivation
- ✓ Protected Nonce Storage
- ✓ Session Memory (Reboot-Cleared)
- ✓ File & Folder Encryption
- ✓ Right-Click Context Menu

How to use this edition

The Standard edition protects files with a Password and an Unlock Code. Both must match exactly to decrypt — if either is lost, files cannot be recovered.

1. Enter your Password in the PASSWORD field. Click SHOW to verify it, CLEAR to reset the session.
2. Enter your Code in the CODE field (e.g. 20.3.3 or 1313).
3. Encryption Mode shows Standard. Click the upgrade link to move to Advanced edition.
4. Choose your Protection Level. Stellare Level 2 is recommended — it applies 7 Stellare layers plus triple-pass AES-256 to ChaCha20 to AES-256.
5. Click ENCRYPT FILE to select and encrypt a single file, or ENCRYPT FOLDER to process an entire folder recursively.
6. The encrypted file is saved with a .sf extension. The original is securely deleted after encryption.
7. Tick Remember this session to save credentials until the next reboot. Untick at any time to clear immediately.

IMPORTANT Use the exact same Password and Code to decrypt. There is no recovery mechanism if either is lost.

B.2 · Advanced Edition

ADVANCED SECURE MODE — KEY FILE · TRIPLE-PASS · 5-KEY DERIVATION

Advanced Edition main dashboard

ADVANCED · £29

- ✓ Everything in Standard +
- ✓ Second Code Layer
- ✓ Simple Key File (.stkf)
- ✓ Triple-Pass Encryption
- ✓ AES-256 → ChaCha20 → AES-256
- ✓ Key-File-Bound Nonce Protection
- ✓ KFM Salt Mixing (Argon2id)
- ✓ 5-Key Independent Derivation

How to use this edition

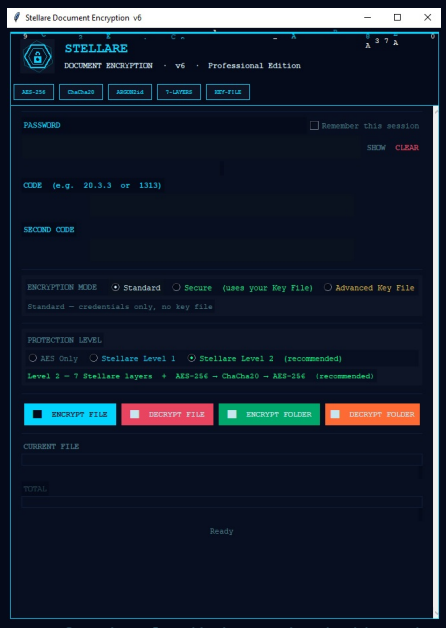
The Advanced edition adds a Second Code and a physical Key File (.stkf) as a third authentication factor. All three must be present to decrypt.

1. Enter your Password, Code, and Second Code in their respective fields.
2. Set Encryption Mode to Secure (Key File) to enable key file protection.
3. Click NEW BLOB to generate a fresh 64KB random .stkf key file, or BROWSE .stkf to select an existing one.
4. Choose Protection Level. Stellare Level 2 applies the full triple-pass cipher: AES-256, then ChaCha20, then AES-256.
5. Click ENCRYPT FILE or ENCRYPT FOLDER. The key file changes the Argon2id salt — without it, decryption is impossible even with correct credentials.
6. The .stkf filename is never stored in the encrypted file — an attacker cannot determine which file was used.

CRITICAL Back up your .stkf file immediately. Losing it makes all files encrypted with it permanently inaccessible.

B.3 · Professional Edition

PRO BRUTE FORCE SECURE — ADVANCED KEY FILE · KFM DEPTH · FORENSIC RESISTANCE



Professional Edition main dashboard

PROFESSIONAL · £59

- ✓ Everything in Std & Adv +
- ✓ Advanced Key File System
- ✓ KFM-Controlled Encryption Depth
- ✓ Defaced Hash Verification
- ✓ Unnamed Island Security
- ✓ Forensic Resistance
- ✓ Anti Brute-Force Architecture
- ✓ Machine-Locked License Binding

How to use this edition

The Professional edition uses any existing file on your machine as the key source. The encryption depth itself is controlled by the key file — an attacker cannot know how many passes were applied without it.

1. Enter your Password, Code, and Second Code.
2. Set Encryption Mode to Advanced Key File.
3. Click BROWSE to nominate any existing file on your machine as the key source. A large file such as a photo or document works best.
4. When Advanced Key File and Stellare Level 2 are both active, the PRO BRUTE FORCE SECURE — ACTIVE banner lights up green at the top of the dashboard.
5. The nominated file's internal byte positions are derived from your password — an attacker with the file still cannot determine which bytes were used.
6. The hash in the encrypted file is deliberately defaced. Forensic analysis cannot confirm the file contains encrypted data.
7. Click ENCRYPT FILE or ENCRYPT FOLDER to protect files with the full Professional stack.

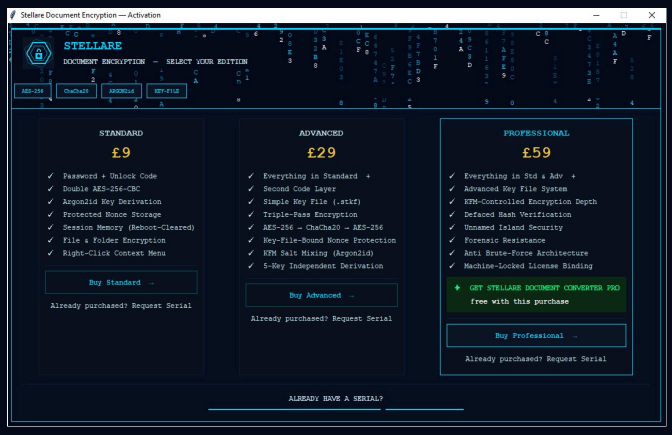
CRITICAL Back up your nominated key file to a separate drive. If it is deleted, modified, or the drive fails, files encrypted with it are permanently inaccessible.

1 · Getting Started

1.1 Activation

Stellare Document Encryption requires a one-time activation tied to your machine. Your licence allows activation on one machine per serial. The activation process is fully automatic — your Machine ID is generated from your hardware and a serial is issued to your email within minutes.

Step 1 of 2 — Request your serial



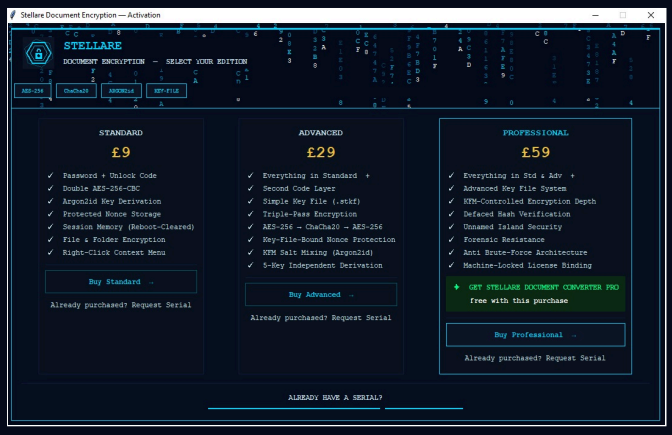
Step 1 of 2 — Send your Machine ID to receive your serial

Click **Already purchased? Request Serial** on your edition card.

Your Machine ID is generated automatically. Enter the email address you used for payment.

Click **OPEN EMAIL CLIENT** — your email client opens pre-filled with your Machine ID. Send it and your serial arrives within minutes.

Step 2 of 2 — Enter your serial



Step 2 of 2 — Enter serial key received by email

Once your serial arrives by email, return to the software.

Paste your serial into the **ALREADY HAVE A SERIAL** field and click **ACTIVATE**.

The serial is validated against your machine hardware. Activation is instant and permanent — no internet connection required after this point.

> NOTE Serial is locked to this machine only. Keep your activation email — it contains your serial key.

1.2 Moving to a New Machine

Each serial is locked to one machine. To activate on a new machine, purchase an additional licence. Contact support at ecoserv301@gmail.com if you need assistance.

2 · Password and Code

Every encryption and decryption operation requires a Password and a Code. Both must match exactly to decrypt. If either is lost, your files cannot be recovered.

2.1 The Password

Your password can be any combination of letters, numbers and symbols. Longer passwords with mixed characters are significantly stronger.

Strength	Example	Notes
Weak	hello	Single word — avoid
Better	mypass99	Short, no symbols
Strong	Str0ng!Pass#2025	Mixed case + symbols
Strongest	My@Dog!sN4me1sMaxwell99	Long + unpredictable

2.2 The Code

The code is an additional credential entered alongside your password. Standard edition uses Password and Code. Advanced and Professional editions add a Second Code as a third independent authentication factor.

CRITICAL You must use the exact same password and code to decrypt that you used to encrypt. There is no recovery mechanism if either is forgotten.

3 · Encryption Tiers

Your edition serial determines which encryption tier is active. The tier is stored in the encrypted file header — decryption is automatic and does not require you to remember which tier was used.

3.1 Standard — Double AES-256-CBC

Password and Code are processed through Argon2id to derive two independent AES keys. Encryption applies two sequential AES-256-CBC passes. Breaking the outer layer reveals only another AES-256 ciphertext with a completely different key — the attack must start over from scratch. A protected nonce is stored with each file, unreadable without the correct credentials.

3.2 Advanced and Professional — Triple Pass

Password, Code and Second Code are processed through Argon2id to derive five independent keys. Encryption applies three passes: AES-256 then ChaCha20 then AES-256. AES and ChaCha20 are from entirely different cryptographic families — a theoretical breakthrough against one provides zero assistance attacking the other. The KFM salt mixing further ensures that the key file changes the salt fed to Argon2id so an attacker who reads the file header still cannot run key derivation correctly.

3.3 Key Derivation — Argon2id

All tiers use Argon2id — the winner of the Password Hashing Competition and the algorithm recommended by OWASP, NIST, and leading security organisations. Each file receives a unique cryptographically random salt. Two identical files encrypted with the same credentials produce completely different ciphertext every time. Rainbow tables and precomputation attacks do not apply.

4 · Key File

The Key File system adds a physical possession factor to your encryption. Without the correct key file on the encrypting machine, decryption is impossible regardless of how many other credentials are known. This is equivalent to a hardware security token.

4.1 Simple Key File (Advanced edition)

The software generates a dedicated 64KB random blob file (.stkf). During encryption, unique positions are derived from the blob using your password+code+salt as the position seed — an attacker cannot determine which bytes were used even with the file. Those bytes extract key material that feeds directly into KFM salt mixing and the Argon2id derivation.

✓ The .stkf file must be backed up — losing it means losing access to encrypted files

✓ The position map is unique per encryption event — same password uses different positions each time

✓ No reference to the .stkf filename is stored anywhere in the encrypted file

4.2 Advanced Key File (Professional edition)

Instead of generating a new file, you nominate any existing file on your machine as the key source. The software derives unique interior positions from the nominated file using your password as the derivation seed — an attacker cannot determine which byte positions were used even if they have the nominated file. KFM-Controlled Encryption Depth means the key file also determines the number and type of extra encryption passes applied, which is unknown to any attacker without the correct key file.

✓ Position derivation seeded from your password — unknown to any attacker

✓ Key material extracted from locked positions feeds directly into Argon2id

✓ The nominated filename is never stored anywhere in the encrypted file

✓ An attacker cannot determine which file on your machine is the key source

4.3 Hash Defacing (Professional edition)

In Professional tier the content verification hash is deliberately incomplete. A portion of the hash bytes are replaced by coordinate data pointing into the key file. The hash cannot be verified, completed or even recognised as a hash without the key file present.

✓ The encrypted file contains no complete verifiable cryptographic structure

✓ Forensic analysis cannot confirm the file contains encrypted data

✓ Plausible deniability at the file format level

CRITICAL Back up your key file immediately after setting it. If the file is deleted, moved or the drive fails, files encrypted with that key file are permanently inaccessible.

5 · Encrypting Files

Main Dashboard

The main dashboard gives you full control over all encryption settings. Professional edition shows the PRO BRUTE FORCE SECURE — ACTIVE status bar when Advanced Key File and Level 2 protection are both enabled.

IMPORTANT The PRO BRUTE FORCE SECURE indicator confirms all maximum protection layers are active simultaneously.

5.1 Single File

✓ Enter your password and code (and second code if Advanced/Professional)
✓ Right-click any file in Explorer and select Encrypt with Stellare
✓ Or open the main application and click ENCRYPT FILE
✓ The encrypted .sf file is created alongside the original

A toast notification confirms each file encrypted.

5.2 Folder Encryption

✓ Right-click any folder in Explorer and select Encrypt Folder with Stellare
✓ All files inside the folder are encrypted recursively
✓ Existing .sf files are automatically skipped
✓ A progress bar shows current file and overall completion

NOTE Original files are not deleted automatically. You may delete them manually once you have verified the .sf files decrypt correctly.

6 · Decrypting Files

Decryption requires the exact same password and code used during encryption. The encryption tier and key file settings are detected automatically from the file header.

6.1 Single File

✓ Double-click any .sf file — the decrypt dialog opens automatically
✓ Or right-click an .sf file and select Decrypt with Stellare
✓ Enter your credentials and click Decrypt

A toast notification confirms each file decrypted.

6.2 Folder Decryption

✓ Right-click a folder in Explorer and select Decrypt Folder with Stellare
✓ All .sf files in the folder are decrypted
✓ Non-.sf files are skipped automatically

NOTE Wrong password or code results in a decryption failure. The .sf file is not damaged — it simply cannot be opened without the correct credentials.

7 · Progress and Cancel

7.1 Progress Bars

During batch folder operations, two progress bars track the operation:

Bar	Colour	Shows
Current File	Cyan	Active filename — resets per file
Total	Green	Overall completion percentage

7.2 Cancel

A STOP button appears during any batch operation. Clicking Stop cancels after the current file completes — it will not interrupt or corrupt a file mid-encryption.

8 · Session Memory

The Remember until reboot checkbox saves your credentials between sessions so you do not need to retype them after each operation.

- ✓ Credentials are saved encrypted using AES-256 locked to your machine
- ✓ The session file is stored in your Windows TEMP folder — automatically cleared on reboot
- ✓ The session file cannot be read on any other machine
- ✓ Key file path is saved alongside credentials for seamless context-menu operation
- ✓ Uncheck the box at any time to clear the saved session immediately

NOTE The session file is machine-locked and AES-256 encrypted. It cannot be read or reused on any other machine.

9 · Security Overview

9.1 Argon2id Key Derivation

All editions use Argon2id — the winner of the Password Hashing Competition and the algorithm recommended by OWASP, NIST, and leading security organisations. Argon2id is memory-hard, meaning GPU and ASIC acceleration provides minimal advantage. Brute-force attacks are computationally expensive regardless of the hardware deployed against it.

9.2 AES-256-CBC

Standard edition applies two independent AES-256-CBC passes. Each pass uses a separate key derived from the Argon2id output. Breaking the outer layer reveals only a second AES-256 ciphertext with a different key — the attack must restart completely.

9.3 ChaCha20

Advanced and Professional editions add a ChaCha20 stream cipher pass between the two AES passes. ChaCha20 was designed by Daniel J. Bernstein and is used in TLS 1.3, WireGuard VPN, and Google's internal infrastructure. AES and ChaCha20 are from entirely different cryptographic families — a theoretical breakthrough against one does not assist an attack against the other.

9.4 5-Key Independent Derivation

Advanced and Professional editions derive five independent 32-byte keys from a single 160-byte Argon2id output. Each key serves a distinct purpose: outer AES, inner AES, ChaCha20, nonce protection, and layer behaviour seeding. No key can be derived from knowledge of another.

9.5 KFM Salt Mixing

When a key file is present, the salt fed to Argon2id is mixed with key file material using a password-seeded XOR pattern. An attacker who reads the file salt from the header still cannot run key derivation correctly without the key file. The actual salt is never stored — it is reconstructed from file salt + key file + password.

9.6 Per-File Random Salt

Every encrypted file receives a unique cryptographically random salt generated at encryption time. Two identical files encrypted with the same credentials produce completely different ciphertext. Rainbow table attacks, precomputation attacks and known-plaintext attacks are not applicable.

9.7 Licence Security

Your activation serial is derived from your machine hardware fingerprint. Each machine produces a unique serial. The licence file is encrypted with a machine-locked key — it cannot be read or reused on any other machine.

10 · Encrypted File Format

Encrypted files use the .sf (Stellare File) extension. The SF6K header contains all metadata required for decryption — you never need to remember settings manually.

Field	Contents
Magic	SF6K — identifies the file format
Version	Format version number
Auth Tier	STD or ADV — which credential tier was used
Key File Tier	None, Simple or Advanced
File Salt	Unique random salt for this file
Hash	Complete (Std/Adv) or defaced (Pro — requires key file to verify)
Coordinate Map	Key file coordinate map — embedded if key file was used
Ciphertext	The encrypted file data

11 · Windows Context Menu

Stellare integrates with the Windows right-click context menu so you can encrypt and decrypt without opening the main application. Context menu entries are installed automatically when you install Stellare.

Right-Click Actions

Action	Result
Right-click any file	Encrypt with Stellare
Right-click any .sf file	Decrypt with Stellare
Right-click any folder	Encrypt / Decrypt Folder
Double-click .sf file	Opens decrypt automatically

NOTE If right-click options are missing after installation, re-run the installer. If a session is active, context menu operations run silently using your saved credentials.

12 · Best Practices

Always Do These

✓ Keep backups of important files before encrypting
✓ Store your password and code in a reputable password manager
✓ Back up your .stkf key file immediately after generating it
✓ Back up your nominated key file (Professional) to a separate drive
✓ Keep your activation email — it contains your serial number
✓ Test decryption on a non-critical file after setting new credentials
✓ Use Professional edition with Advanced Key File for maximum sensitivity files

Never Do These

✗ Never encrypt without keeping a backup
✗ Never forget your password or code — there is no recovery
✗ Never delete or move your key file while it is active
✗ Never use this software to store or conceal unlawful material

13 · Troubleshooting

Problem	Solution
Invalid serial key	Check the serial is copied correctly with all dashes. Use the email you purchased with. Contact support if the issue persists.
Decryption fails	Ensure password and code are exactly as used during encryption. Check for accidental spaces or capitalisation.
Key file warning	The saved key file cannot be found. Restore it to its original location or browse to its new location.
File not found after decrypt	The original file is restored to the same folder as the .sf file.
Activation window after reinstall	Reactivate using your existing serial from the purchase email.
Email not received	Check spam/junk. Wait up to 10 minutes. Use Already purchased to request a resend.
Serial rejected on new machine	Each serial is locked to one machine. Contact support to transfer.
Context menu missing	Re-run the installer and ensure context menu integration is selected.

If your issue is not listed above, contact support at ecoserv301@gmail.com

14 · Legal and EULA

14.1 Licence Grant

Stellare Software grants you a non-exclusive, non-transferable personal licence to use Stellare Document Encryption on one machine per serial purchased. Additional machines require purchase of additional licences.

14.2 No Warranty

This software is provided AS IS without warranty of any kind. Stellare Software makes no warranty that the software will be error-free or meet your specific requirements.

14.3 Data Loss Warning

Stellare Software accepts no liability for data loss resulting from use of this software. You must maintain backups of all important files.

14.4 No Key Recovery

Stellare Software has no access to your passwords, codes, key files or the contents of your encrypted files at any time. We cannot recover your data if credentials are lost.

14.5 Lawful Use

You agree to use this software strictly for lawful purposes in compliance with the laws of your country.

14.6 Governing Law

This agreement is governed by the laws of England and Wales.

Developer	Stellare Software
Support	ecoserv301@gmail.com
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